Problem 4: Epun (25 marks)

Here are some sentences in Epun[8] and their English translations:[9]

Midebiniz sminus fa vakvepiz.	Some kings kiss the walruses$_M$.
Chavetius f imnikiniz gub midep.	The cats$_M$ won't love this queen.
Tak huspat asa smin?	What have they$_M$ kissed?
Asa hahielogus chegodiniz.	Some dogs$_F$ haven't looked at them.
Guza hakakolgu gub zenadin.	This girl hasn't seen her.
Chupanibu fa pelikin.	The turtle$_M$ won't come.
Osozeu uza imnipiz.	He doesn't catch some cats$_F$.
Tak hagupat gub imnidin kakol?	What has this cat$_F$ seen?
Afa bozudin cherehelgu gub vletop.	The teacher$_F$ will write this letter.
Tak chupat f imnikin osoze?	What will the cat$_M$ catch?
Nisa osozenis fa chegodopiz.	We catch the dogs$_M$.
Af imnipiz bozudiniz havetigus.	Some teachers$_F$ have loved the cats$_F$.
Zaddiliniz hapanibus.	Some men have come.
Makohin binapu.	The boy returns.

Assignment 1: (3 marks)

Translate into English:

(a) Tak hugupat panib?

(b) Uza havetigu guza.

(c) Pelikop nisa hakakolnis.

Assignment 2: (8 marks)

Translate into Epun:

(a) The walrus$_F$ will kiss us.

(b) This man hasn't written a letter.

(c) Some turtles$_F$ don't look at a woman.

(d) The boys have loved the queens.

Assignment 3: (6 marks)

Translate into Epun in all possible ways:

(a) What do they love?

[8] Epun is an invented language, deliberately devised to contain constructions which violated universal grammatical principles.

[9] ch = ch in chop. The subscripts M and F refer to masculine and feminine, respectively.

(b) What will return?

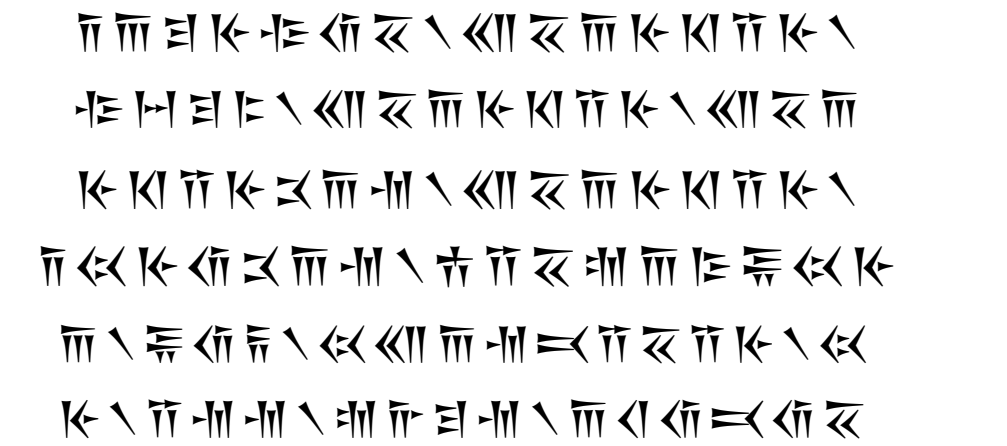
Assignment 4: (8 marks)
Explain your solution.

– Vlad A. Neacşu (inspired by a problem created by Adina Dragomirescu)

Problem 5: Old Persian (20 marks)

Old Persian was spoken from 600-300 BCE in Achaemenid Persia. It was written in cuneiform[10]. Since the medieval times travelers had noticed the Old Persian inscriptions in cuneiform carved in the palaces and ruins of Persepolis, a city in Iran. However, the script remained undeciphered till 18th century. In 1802, Georg Friedrich Grotefend, a German philologist, managed to (almost correctly) decipher most of the script starting from two simple inscriptions, known as the Niebuhr inscriptions 1 and 2.[11]

Niebuhr inscription 1:



Niebuhr Inscription 2:



Based on the repetition patterns in the two inscriptions and context in which they occur, Grotefend correctly guessed that the translation of the first inscription should be:

[10] Cuneiform was a writing system produced using wedge-shaped marks in clay tablets, inherited from the Sumerian Empire.

[11] Slightly simplified for this problem.